Huawei Access Node HEB-ACC-40 Hebbal Edge QoS Congestion Event SOP

Node: HEB-ACC-40 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_1431

Description

This Standard Operating Procedure (SOP) details the diagnostic and resolution process for a Quality of Service (QoS) congestion event on Huawei Access Node HEB-ACC-40 located at Hebbal Edge. The failure domain occurs due to microbursts, interface tail drops, and QoS queue starvation. The underlying protocol and hardware mechanics causing this failure domain are rooted in the Hierarchical QoS (HQoS) and the Multi-Field Classification (MFC) mechanism. The HQoS configuration on the Huawei Access Node enables differentiated services for various traffic classes, but if not properly configured or if there are hardware limitations, it can lead to congestion and packet drops. The MFC mechanism classifies packets based on multiple fields such as source and destination IP addresses, ports, and protocols, but if the classification rules are not correctly defined or if the hardware resources are insufficient, it can result in QoS queue starvation.

Pre-requisite Checks

- Ensure remote hands are dispatched to Hebbal Edge for physical access to the Huawei Access Node.

- Verify the status of the redundant link at Hebbal Edge to ensure it is operational.

- Check the power supply and environmental conditions of the Huawei Access Node to rule out any power-related issues.

Potential Root Causes

Cause	Probability
Incorrect HQoS configuration	High
Insufficient hardware resources for MFC	Medium
Misconfigured classification rules	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos profile, Expected Output: QoS profile information, including classification rules and queue configurations, Description: This command displays the current HQoS profile configuration, including classification rules and queue configurations. It helps identify any misconfigured rules or queue settings that may be contributing to the congestion event.

Command: display interface queue, Expected Output: Interface queue information, including packet drops and queue lengths, Description: This command displays the current interface queue information, including packet drops and queue lengths. It helps identify any interface-related issues that may be contributing to the congestion
```

```
event.

Command: display mfc rule, Expected Output: MFC rule information, including classification rules and match criteria, Description: This command displays the current MFC rule configuration, including classification rules and match criteria. It helps identify any misconfigured rules or match criteria that may be contributing to the congestion event.
```

Resolution Steps

Step 1: Step: Verify HQoS configuration, Command: display qos profile, Description: Verify the current HQoS profile configuration to ensure it is correctly defined and meets the traffic requirements.

Step 2: Step: Adjust HQoS configuration, Command: qos profile profile-name modify, Description: Adjust the HQoS profile configuration as needed to ensure proper classification and queueing of traffic.

Step 3: Step: Verify MFC configuration, Command: display mfc rule, Description: Verify the current MFC rule configuration to ensure it is correctly defined and meets the traffic requirements.

Step 4: Step: Adjust MFC configuration, Command: mfc rule rule-name modify, Description: Adjust the MFC rule configuration as needed to ensure proper classification and queuing of traffic.

Post-Check Verification

```
Command: display qos profile, Expected Output: QoS profile information, including classification rules and queue configurations, Description: Verify the HQoS profile configuration after adjustments to ensure it is correctly defined and meets the traffic requirements.

Command: display interface queue, Expected Output: Interface queue information, including packet drops and queue lengths, Description: Verify the interface queue information after adjustments to ensure it is correctly configured and meets the traffic requirements.

Command: display mfc rule, Expected Output: MFC rule information, including classification rules and match criteria, Description: Verify the MFC rule configuration after adjustments to ensure it is correctly defined and meets the traffic requirements.
```

Roll-back Procedures

- Step: Restore original HQoS configuration, Command: qos profile profile-name restore, Description: Restore the original HQoS profile configuration if the adjustments did not resolve the issue.

- Step: Restore original MFC configuration, Command: mfc rule rule-name restore, Description: Restore the original MFC rule configuration if the adjustments did not resolve the issue.